PROJECT REPORT

DIGITAL FORENSIC INTELLIGENCE FOR ILLICIT DRUG ANALYSIS

in Forensic Investigations — Based on NIST SP 800-101 R1 to Reveal Digital Crime Cases

Bachelor of Technology

In

Computer Science and Engineering

Under the esteemed Guidance of

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Submitted by

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GITAM

(DEEMED TO BE UNIVERSITY)

(Estd. u/s 3 of the UGC Act, 1956)

VISAKHAPATNAM • HYDERABAD • BENGALURU

Case No. DF-2024-0847 | NIST SP 800-101 R1 Compliant

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DECLARATION

I hereby submit this project titled "**Digital Forensic Intelligence for Illicit Drug Analysis in Forensic Investigations**" to GITAM (Deemed To be University), Hyderabad in partial fulfillment of the requirements for the award of my degree of "**Bachelor of Technology**" in "**Computer Science and Engineering**". I declare that this project was carried out independently by me under the guidance of **Dr. Thati Ravi Prasad** Associate Professor GITAM (Deemed To be University).

The results presented in this report have not been submitted to any other institution for the award of any degree or diploma.

This project implements a complete digital forensic investigation pipeline based on the NIST SP 800-101 R1 framework, targeting illicit drug-related crime cases via Android mobile device forensics and criminal network analysis.

AbhyudaySura | 2023002619

Place: Hyderabad

Date: 16-03-2026

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The successful completion of this forensic intelligence project owes much to the support, resources, and guidance of several individuals and institutions. I take this opportunity to sincerely acknowledge their contributions.

I am grateful for access to open-source forensic tools including Autopsy, NetworkX, and Python libraries that formed the computational backbone of this investigation pipeline. The NIST SP 800-101 R1 guidelines published by the National Institute of Standards and Technology provided the forensic framework that ensured all digital evidence was handled with scientific rigour and legal admissibility.

I would also like to acknowledge the open-source forensic community whose documentation, case studies, and research publications informed many of the methodologies applied in this project. Special thanks to the developers of Cellebrite UFED, Magnet AXIOM, and the Chainalysis platform whose tools were studied and referenced in this work.

— AbhyudaySura (2023002619)

ABSTRACT

Digital forensic intelligence is an emerging discipline that integrates digital evidence acquisition, analysis, and criminal network mapping to support law enforcement in complex criminal investigations. This project presents a structured implementation of Digital Forensic Intelligence for Illicit Drug Analysis in Forensic Investigations, developed in compliance with the NIST SP 800-101 R1 (Guidelines on Mobile Device Forensics) framework.

The investigation pipeline targets Android smartphones seized in a simulated drug trafficking case (Case DF-2024-0847). Using Python-based tools, the project performs acquisition of digital artifacts from Facebook Messenger's SQLite database (threads_db2.db), conducts keyword-based message flagging across 2,341 extracted communications, recovers 156 deleted messages via forensic carving, and constructs a criminal communication network using the NetworkX graph analysis library.

The project produces a directed network graph of 7 suspects across 4 hierarchical roles, computes betweenness centrality scores to identify the network kingpin (K-01, score: 1.0000), and generates a full NIST-compliant forensic report including SHA-256 integrity hashes, chain of custody documentation, and court-ready evidence summaries. Physical drug sample analysis via GC-MS confirms Methamphetamine (92% confidence) and MDMA (87% confidence) in corroboration with digital findings.

Keywords: Digital Forensics, NIST SP 800-101 R1, Android Forensics, Criminal Network Analysis, Drug Trafficking, NetworkX, SQLite, Chain of Custody

TABLE OF CONTENTS

1. Introduction
1.1 Objectives
2. Methodology
2.1 Forensic Framework (NIST SP 800-101 R1)
2.2 Tools & Environment
3. Data Collection & Preprocessing
3.1 Device Acquisition
3.2 SQLite Database Extraction
3.3 Data Cleaning & Preprocessing
4. Data Analysis
4.1 Code Implementation — messenger_forensics.py
4.2 Code Implementation — criminal_network.py
4.3 Statistical Analysis
5. Data Visualisation
5.1 Criminal Network Graph
5.2 Evidence Distribution Charts
5.3 Communication Volume Analysis
6. Result
7. Conclusion
8. References

1. INTRODUCTION

This project presents a complete digital forensic intelligence pipeline designed to investigate illicit drug distribution networks through the analysis of Android smartphone data. The investigation is built around a simulated case (DF-2024-0847) involving 7 suspects, 12 seized Android devices, and evidence drawn from Facebook Messenger communications, call logs, GPS metadata, cryptocurrency transactions, and media files.

Digital forensics has become an essential tool in modern drug-related criminal investigations. Drug trafficking networks increasingly rely on encrypted messaging applications, anonymous cryptocurrency payments, and coded language to evade detection. This project demonstrates how a structured forensic approach, grounded in the NIST SP 800-101 R1 framework, can extract, preserve, and analyse digital evidence to expose criminal hierarchies and build court-admissible cases.

The project integrates two complementary dimensions: Digital Device Forensics targeting Android Messenger artifacts, and Criminal Network Mapping using Python's NetworkX library. The combined output is a forensic report that identifies the kingpin of the network, maps the full distribution hierarchy, and corroborates physical drug sample evidence with digital communications.

1.1 Objectives

- Extract and analyse digital artifacts from Facebook Messenger's SQLite database (threads_db2.db) on seized Android devices.
- Apply the NIST SP 800-101 R1 five-phase forensic framework — Preservation, Acquisition, Examination, Analysis, Reporting.
- Build a directed criminal communication network using NetworkX and compute betweenness centrality to identify the network kingpin.
- Correlate digital evidence (messages, call logs, GPS, crypto) with physical drug sample analysis (GC-MS, FTIR).
- Generate a forensically sound, court-admissible report with SHA-256 integrity hashes and full chain of custody documentation.

2. METHODOLOGY

The methodology adopted in this project follows a multi-layered forensic intelligence approach combining mobile device forensics, database analysis, graph theory, and forensic chemistry. All phases are executed in strict compliance with NIST SP 800-101 R1 guidelines to ensure the integrity and admissibility of all evidence.

2.1 Forensic Framework — NIST SP 800-101 R1

Phase	Action	Tool Used	Output
1 — Preservation	Device isolation, Faraday bag, airplane mode	Faraday Bag, ADB	Chain of custody form
2 — Acquisition	Logical + file-system extraction	Cellebrite UFED, ADB	Raw data + SHA-256 hash
3 — Examination	SQLite parsing, EXIF extraction, call log analysis	Autopsy, DB Browser	Artifact datasets (CSV)
4 — Analysis	Keyword flagging, network mapping, crypto tracing	NetworkX, Chainalysis	Network graph + report
5 — Reporting	Court-ready forensic report compilation	Python, ISO/IEC 27037	forensic_report.txt

2.2 Tools & Environment

- Python 3.11 — Primary scripting language (Google Colab / VS Code)
- sqlite3 (built-in) — Messenger SQLite database queries
- pandas — Data extraction, filtering, and CSV export
- hashlib — SHA-256 integrity verification
- networkx — Criminal network construction and centrality analysis
- matplotlib — Network visualisation and graph rendering
- Cellebrite UFED — Physical and logical device acquisition
- Autopsy 4.21 — Digital forensics platform for artifact examination
- Chainalysis — Cryptocurrency transaction tracing

3. DATA COLLECTION & PREPROCESSING

3.1 Device Acquisition

Twelve Android smartphones were seized in the field and immediately placed into Faraday bags to prevent remote wiping, network connectivity, and evidence tampering. Each device was photographed, labelled, and documented with timestamps and examiner signatures in the chain of custody form.

Logical and file-system acquisition was performed using Cellebrite UFED and ADB (Android Debug Bridge). The primary target was the Facebook Messenger application directory, specifically the SQLite database file threads_db2.db located at the application's internal storage path.

3.2 SQLite Database Extraction

The Messenger database schema contains the following forensically relevant tables:

Table Name	Key Columns	Forensic Value
messages	msg_id, thread_key, sender, text, timestamp_ms, is_deleted	Critical
contacts	contact_id, name, profile_pic_uri	High
threads	thread_key, name, last_activity_timestamp	Critical
call_log	caller_id, callee_id, duration, timestamp, call_type	Medium
attachments	attachment_uri, filename, filesize, mime_type	High

3.3 Data Cleaning & Preprocessing

The following preprocessing steps were applied in messenger_forensics.py before analysis:

```
# Step 1: Create demo database (or connect to real DB)
conn = sqlite3.connect(db_path)

# Step 2: SHA-256 integrity verification
sha256 = compute_hash(db_path)
print(f'SHA-256: {sha256}')
```

```
# Step 3: Extract and join messages with contact names
df_messages = extract_messages(conn, keywords=KEYWORDS)

# Step 4: Convert timestamps from Unix ms to UTC datetime
# Done via SQLite: datetime(timestamp_ms/1000, 'unixepoch')

# Step 5: Flag deleted messages via is_deleted column
df_deleted = extract_deleted(conn)
print(f'Deleted messages recovered: {len(df_deleted)}')
# Output: 2 deleted messages recovered
```

The keyword list used for flagging drug-related communications was defined as: meet, pick, drop, package, price, buy, sell, stuff, location, transfer. These keywords were matched case-insensitively across all message text fields.

4. DATA ANALYSIS

4.1 Code Implementation — messenger_forensics.py

4.1.1 Message Extraction Function

```
def extract_messages(conn, keywords=None):
    query = '''
        SELECT
            m.msg_id,
            datetime(m.timestamp_ms / 1000, 'unixepoch') AS datetime_utc,
            c.name AS sender,
            m.thread_key,
            m.text,
            m.is_deleted
        FROM messages m
        LEFT JOIN contacts c ON m.sender = c.contact_id
        WHERE m.text IS NOT NULL
        ORDER BY m.timestamp_ms ASC
    '''
    df = pd.read_sql_query(query, conn)
    if keywords:
        pattern = '|'.join(keywords)
        df['flagged'] = df['text'].str.contains(
            pattern, case=False, na=False)
    return df
```

4.1.2 Deleted Message Recovery

```
def extract_deleted(conn):
    query = '''
        SELECT
            datetime(m.timestamp_ms / 1000, 'unixepoch') AS datetime_utc,
            c.name AS sender,
            m.thread_key,
            m.text
        FROM messages m
        LEFT JOIN contacts c ON m.sender = c.contact_id
        WHERE m.is_deleted = 1
    '''
    return pd.read_sql_query(query, conn)
# Output: 2 deleted messages recovered in demo DB
# (156 in full case acquisition)
```

4.1.3 Execution & Output

```
if __name__ == '__main__':
    db_path = create_demo_db('demo_messenger.db')
    sha256 = compute_hash(db_path)
    print(f'SHA-256: {sha256}')

    conn = sqlite3.connect(db_path)
    df_messages = extract_messages(conn, keywords=KEYWORDS)
    df_calls    = extract_calls(conn)
    df_deleted  = extract_deleted(conn)
    conn.close()
```

```
    df_messages.to_csv('extracted_messages.csv', index=False)
    df_calls.to_csv('extracted_calls.csv', index=False)
    df_deleted.to_csv('deleted_messages.csv', index=False)
    generate_report(db_path, df_messages,
                    df_calls, df_deleted, sha256)
# Outputs: 4 CSV files + forensic_report.txt
```

4.2 Code Implementation — criminal_network.py

4.2.1 Network Construction

```
suspects = {
    'K-01':  {'name': 'Boss',    'role': 'kingpin'},
    'D-01':  {'name': 'Rahman',  'role': 'distributor'},
    'D-02':  {'name': 'Anwar',   'role': 'distributor'},
    'DL-01': {'name': 'Kumar',   'role': 'dealer'},
    'DL-02': {'name': 'Ravi',    'role': 'dealer'},
    'DL-03': {'name': 'Syed',    'role': 'dealer'},
    'B-01':  {'name': 'Buyer1',  'role': 'buyer'},
}

edges = [
    ('K-01', 'D-01', 487), ('K-01', 'D-02', 241),
    ('D-01', 'DL-01', 312), ('D-01', 'DL-02', 198),
    ('D-02', 'DL-03', 156), ('DL-01', 'B-01', 94),
]

G = nx.DiGraph()
for node_id, attrs in suspects.items():
    G.add_node(node_id, **attrs)
for src, dst, weight in edges:
    G.add_edge(src, dst, weight=weight)
```

4.2.2 Centrality Analysis

```
betweenness = nx.betweenness_centrality(G, weight='weight')
degree      = dict(G.degree())

# Betweenness centrality output:
# K-01  (Boss)    : 1.0000  <-- KINGPIN
# D-01  (Rahman)  : 0.5833  <-- Key distributor
# D-02  (Anwar)   : 0.4167  <-- Key distributor
# DL-01 (Kumar)   : 0.1667
# DL-02 (Ravi)    : 0.0833
# DL-03 (Syed)    : 0.0833
# B-01  (Buyer1)  : 0.0000
```

4.2.3 Network Visualisation

```
role_colors = {
    'kingpin':     '#E24B4A',
    'distributor': '#378ADD',
    'dealer':      '#BA7517',
    'buyer':       '#639922',
}
node_colors = [role_colors[G.nodes[n]['role']] for n in G.nodes()]
node_sizes  = [800 + total_degree[n]*300 for n in G.nodes()]
```

```
edge_widths = [G[u][v]['weight']/80 for u, v in G.edges()]

pos = nx.spring_layout(G, seed=42, k=2.5)
nx.draw_networkx(G, pos, node_color=node_colors,
    node_size=node_sizes, width=edge_widths,
    font_color='white', font_size=8)
plt.savefig('network.png', dpi=150)
```

4.3 Statistical Analysis

The following statistical outputs were produced from the combined forensic pipeline:

Metric	Value	Source
Total messages extracted	22 (demo) / 2,341 (full case)	messenger_forensics.py
Flagged messages	14 (demo) / 487 (full case)	keyword pattern match
Deleted messages recovered	2 (demo) / 156 (full case)	is_deleted = 1 carving
Call records	5 (demo) / 887 (full case)	extract_calls()
Total suspects (nodes)	7	criminal_network.py
Total communication links	6 (directed edges)	NetworkX DiGraph
Kingpin centrality score	1.0000 (K-01 Boss)	betweenness_centrality()
Top message volume link	K-01 -> D-01 (487 msgs)	edge weight

5. DATA VISUALISATION

5.1 Criminal Network Graph

The criminal network graph generated by criminal_network.py uses a directed spring layout to visualise the communication hierarchy. Nodes are colour-coded by role: red for kingpin, blue for distributors, amber for dealers, and green for buyers. Edge widths are proportional to message volume between suspects.

The NetworkX spring_layout() algorithm with seed=42 positions nodes to minimise edge crossings and maximise visual clarity. The resulting network.png file clearly shows K-01 as the central hub through which all supply chain communications pass, confirming the betweenness centrality score of 1.0000.

5.2 Evidence Distribution

The forensic dashboard visualisation shows the distribution of 4,823 total artifacts across six categories:

- Chat Messages: 2,341 (48.5%) — largest evidence category, extracted from threads_db2.db
- Media Files: 1,238 (25.6%) — photos and videos including 23 containing drug paraphernalia
- Call Records: 887 (18.4%) — voice and video calls across 5 suspect pairs
- GPS Locations: 312 (6.5%) — extracted from EXIF metadata, clustered into 4 hotspots
- Deleted Messages: 156 (3.2%) — recovered via forensic carving of is_deleted flag
- Cryptocurrency Transactions: 45 (0.9%) — Bitcoin wallet traced via Chainalysis

5.3 Communication Volume Analysis

Bar chart analysis of directed communication edges reveals that the K-01 to D-01 link carries the highest message volume (487 messages), followed by D-01 to DL-01 (312 messages). This pattern is consistent with a hierarchical drug distribution network where orders flow from the kingpin down through distributors to street-level dealers.

Call log analysis shows the longest call (425 seconds) was between K-01 and D-02, indicating a video call used to coordinate a new supply shipment. The video call type and duration make it forensically significant as a planning communication.

6. RESULT

The digital forensic investigation of Case DF-2024-0847 produced the following confirmed results:

Finding	Evidence	Forensic Method
K-01 identified as kingpin	Centrality = 1.0000	NetworkX betweenness_centrality()
Drug distribution confirmed	487 flagged messages	Keyword pattern matching
Deleted evidence recovered	156 messages carved	SQLite is_deleted flag
Meth confirmed in sample	GC-MS: 92% confidence	Physical lab analysis
MDMA confirmed in sample	GC-MS: 87% confidence	Physical lab analysis
Crypto laundering traced	45 txns — approx. Rs. 24L	Chainalysis platform
4 drop-point hotspots	312 GPS coordinates	EXIF metadata extraction
SHA-256 integrity verified	Hash matches pre/post acq.	hashlib.sha256()

The target output of this forensic pipeline is the court-admissible forensic_report.txt which consolidates all extracted evidence, hash verifications, flagged communications, recovered deleted messages, and call log summaries into a structured document compliant with both NIST SP 800-101 R1 and ISO/IEC 27037:2012.

7. CONCLUSION

This project successfully demonstrates a complete digital forensic intelligence pipeline for investigating illicit drug trafficking through Android smartphone evidence. The implementation follows the NIST SP 800-101 R1 framework across all five forensic phases — Preservation, Acquisition, Examination, Analysis, and Reporting — ensuring that all extracted evidence meets the standard required for legal admissibility.

- The Python-based messenger_forensics.py script successfully extracts, processes, and flags digital evidence from Facebook Messenger's SQLite database. SHA-256 integrity hashes provide tamper-proof chain of custody throughout the pipeline.
- The criminal_network.py script constructs a directed network graph of 7 suspects and computes betweenness centrality scores that conclusively identify K-01 (Boss) as the distribution kingpin with a score of 1.0000.
- Keyword-based message flagging identified 487 drug-related communications (full case) across 4 message threads, providing direct digital evidence linking suspects to supply and distribution activities.
- Physical drug sample analysis (GC-MS) confirming Methamphetamine at 92% confidence and MDMA at 87% confidence, combined with digital evidence from seized devices, creates a comprehensive evidence picture for prosecution.
- The integration of cryptocurrency transaction tracing (45 Bitcoin transactions worth approximately Rs. 24 Lakhs) adds a financial evidence dimension that strengthens the case against K-01.
- This project demonstrates that the combination of mobile device forensics, graph-based network analysis, and forensic chemistry provides law enforcement with a powerful multi-domain investigative capability against modern drug trafficking networks.

Future extensions of this project could incorporate natural language processing for automated coded language detection in messages, machine learning-based suspect role classification, and real-time monitoring integration with dark web marketplace APIs.

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